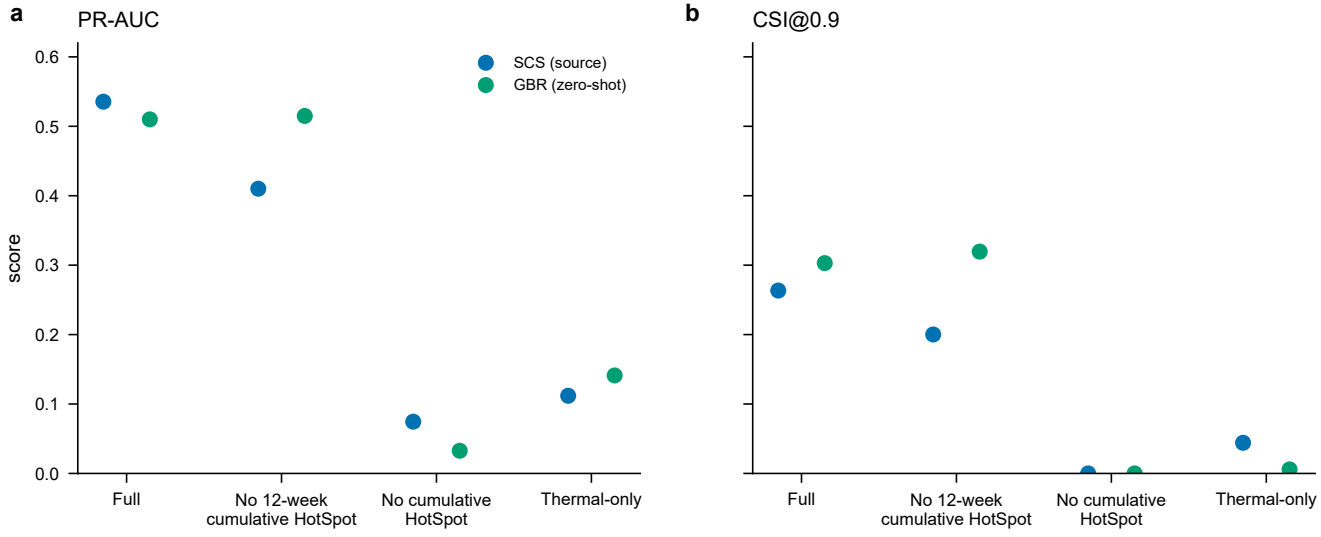



*Retrained feature ablations under the frozen protocol (ocean-masked PFV record sets; SCS n=49,322, GBR n=19,065). Removing the cumulative HotSpot channels eliminates severe-risk discrimination in both basins; the 12-week component is one of two cumulative windows and is not claimed to be universally necessary.*